

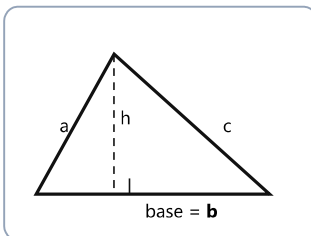
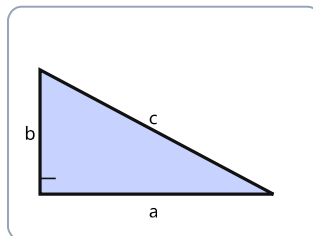
**How to use:** This sheet uses native *MathML* for clean, LaTeX-style math rendering in modern browsers (Chrome, Edge, Firefox, Safari). Click *Print / Export to PDF* to save an A4 PDF for study. Variables are consistently named: **a**, **b**, **c** (sides), **l**, **w**, **h** (length, width, height), **r**, **d** (radius, diameter), **P** (perimeter), **A** (area), **SA** (surface area), **V** (volume), **s** (semi-perimeter or slant height by context).

## Downloadable Formula Sheets

- [Keystone Geometry Formula Sheet](#) — concise reference aligned with Keystone exam topics.
- [M-G2 Geometric Formulas](#) — collection of 2D and 3D formulas with illustrative diagrams.
- [Math Resources Geometry Formulas](#) — general resource covering common geometry formulas.
- [Comprehensive Geometric Formulas](#) — compact sheet summarizing fundamental equations.

## 2D Shapes

### Triangle

Perimeter **P**Area **A**Right triangle hypotenuse **c**Semi-perimeter **s**

- **Perimeter:**

$$P = a + b + c$$

- **Area (with base *b*, height *h*):**

$$A = \frac{1}{2}bh$$

- **Heron's formula:**

Let  $s = \frac{a+b+c}{2}$ . Then

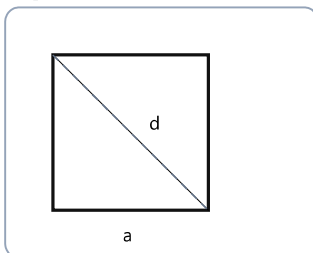
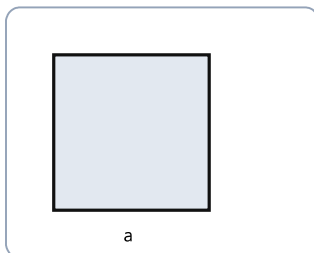
$$A = \sqrt{s(s-a)(s-b)(s-c)}$$

- **Pythagorean theorem (right triangle):**

$$c^2 = a^2 + b^2$$

- **Law of cosines:**

$$c^2 = a^2 + b^2 - 2ab\cos\gamma$$

**Square**side **a**diagonal **d**• **Perimeter:**

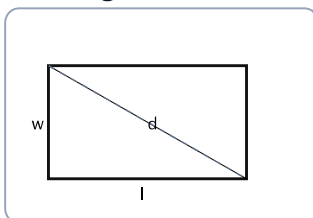
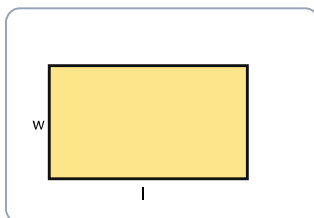
$$P = 4a$$

• **Area:**

$$A = a^2$$

• **Diagonal:**

$$d = a\sqrt{2}$$

**Rectangle**length **l**width **w**diagonal **d**• **Perimeter:**

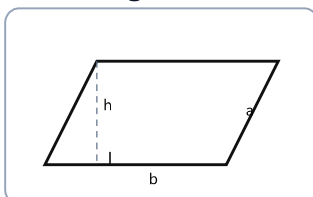
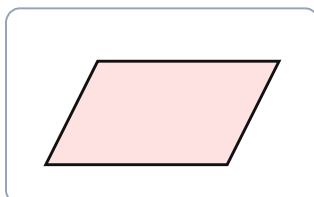
$$P = 2(l + w)$$

• **Area:**

$$A = lw$$

• **Diagonal:**

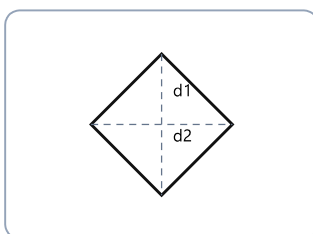
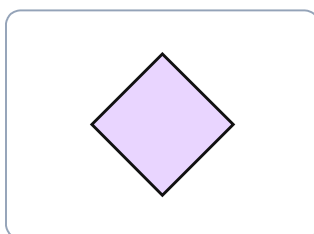
$$d = \sqrt{l^2 + w^2}$$

**Parallelogram**adjacent sides **a, b**height **h**• **Perimeter:**

$$P = 2(a + b)$$

• **Area:**

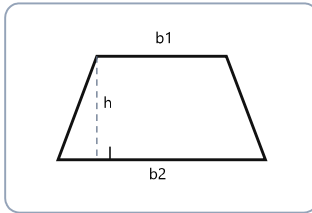
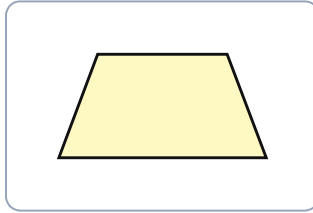
$$A = bh$$

**Rhombus**diagonals **d1, d2**• **Perimeter:**

$$P = 4a$$

(all sides equal to **a**)• **Area:**

$$A = \frac{d_1 \cdot d_2}{2}$$

**Trapezoid (US) / Trapezium (UK)**bases **b1, b2**height **h**

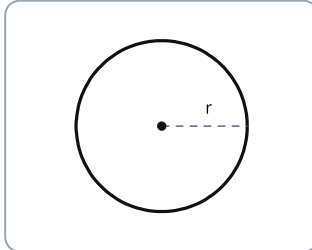
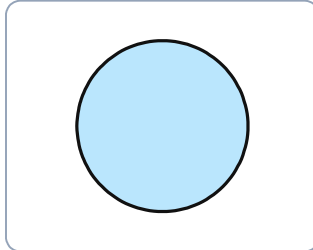
- **Perimeter:** Sum of all side lengths.

- **Area:**

$$A = \frac{b1 + b2}{2} h$$

- **Median length:**

$$m = \frac{b1 + b2}{2}$$

**Circle**radius **r**diameter **d=2r**

- **Circumference:**

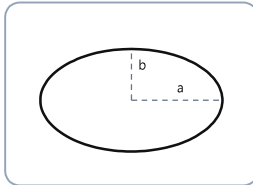
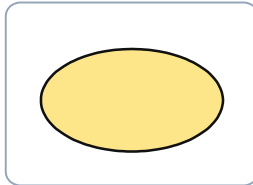
$$C = 2\pi r = \pi d$$

- **Area:**

$$A = \pi r^2$$

- **Sector (angle  $\theta$  in radians):**

$$\text{Arc length } L = \theta r, \text{ Area } A = \frac{\theta}{2} r^2$$

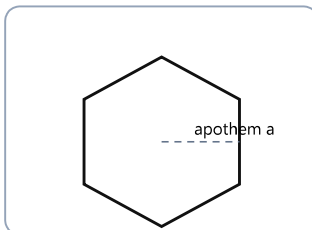
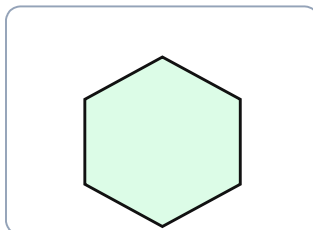
**Ellipse**semi-major **a**semi-minor **b**

- **Area:**

$$A = \pi ab$$

- **Circumference (Ramanujan approx.):**

$$C \approx \pi \left[ 3(a+b) - \sqrt{(3a+b)(a+3b)} \right]$$

**Regular  $n$ -gon**side length **s**apothem **a****n** sides

- **Perimeter:**

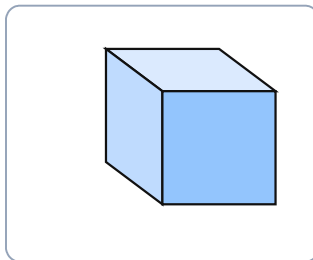
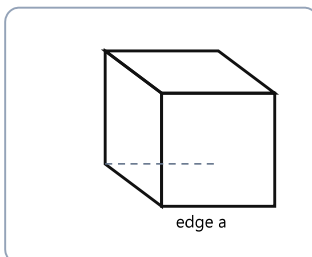
$$P = ns$$

- **Area (two forms):**

$$A = \frac{Pa}{2} = \frac{ns^2}{4 \tan\left(\frac{\pi}{n}\right)}$$

## 3D Shapes

### Cube



edge **a**

- **Surface area:**

$$SA = 6a^2$$

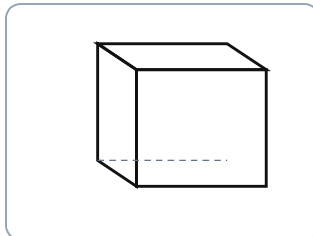
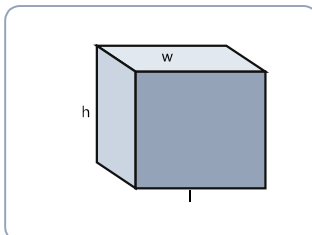
- **Volume:**

$$V = a^3$$

- **Space diagonal:**

$$d = a\sqrt{3}$$

### Rectangular Prism (Cuboid)



length **l**

width **w**

height **h**

- **Surface area:**

$$SA = 2(lw + lh + wh)$$

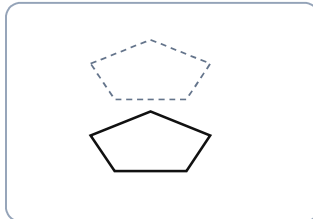
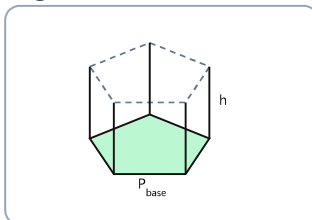
- **Volume:**

$$V = lwh$$

- **Space diagonal:**

$$d = \sqrt{l^2 + w^2 + h^2}$$

### Right Prism (General base)



base area **A<sub>base</sub>**

base perimeter **P<sub>base</sub>**

height **h**

- **Lateral area:**

$$LA = P_{\text{base}}h$$

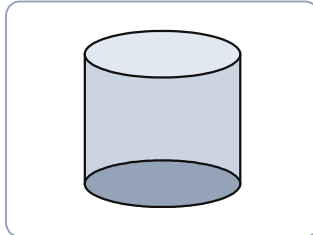
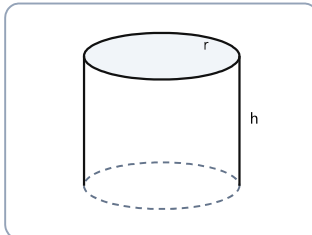
- **Surface area:**

$$SA = LA + 2A_{\text{base}}$$

- **Volume:**

$$V = A_{\text{base}}h$$

### Right Circular Cylinder



radius **r**

height **h**

- **Lateral area:**

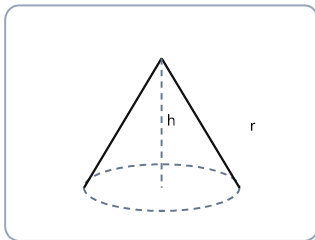
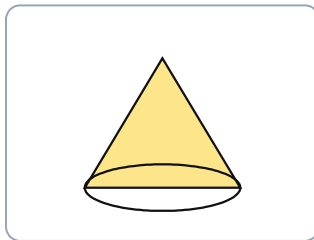
$$LA = 2\pi rh$$

- **Surface area:**

$$SA = 2\pi rh + 2\pi r^2$$

- **Volume:**

$$V = \pi r^2 h$$

**Right Circular Cone**radius **r**height **h**slant **s**

- **Slant height:**

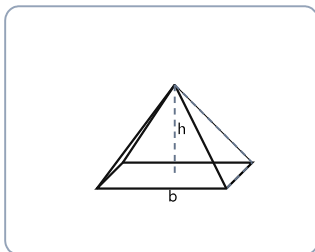
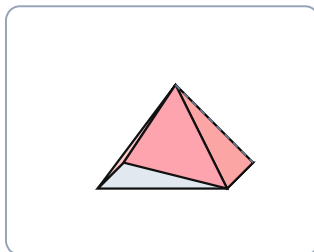
$$s = \sqrt{r^2 + h^2}$$

- **Surface area:**

$$SA = \pi rs + \pi r^2$$

- **Volume:**

$$V = \frac{1}{3}\pi r^2 h$$

**Right Square Pyramid**base side **b**height **h**slant (face) **s**

- **Slant height (face center):**

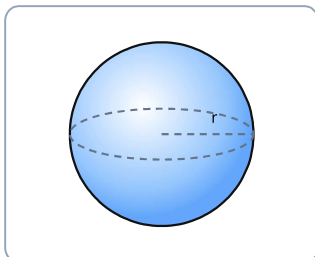
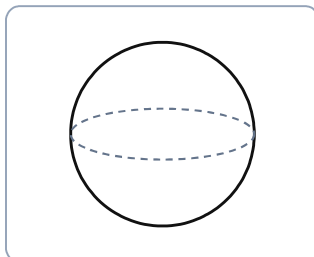
$$s = \sqrt{\frac{b^2}{2} + h^2}$$

- **Surface area:**

$$SA = b^2 + 2bs$$

- **Volume:**

$$V = \frac{1}{3}b^2 h$$

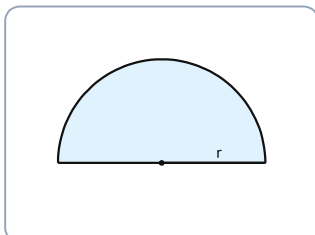
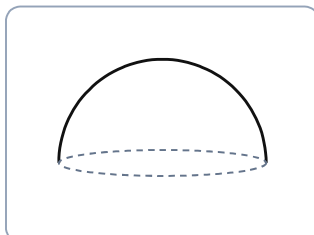
**Sphere**radius **r**

- **Surface area:**

$$SA = 4\pi r^2$$

- **Volume:**

$$V = \frac{4}{3}\pi r^3$$

**Hemisphere**radius **r**

- **Curved surface area (no base):**

$$CSA = 2\pi r^2$$

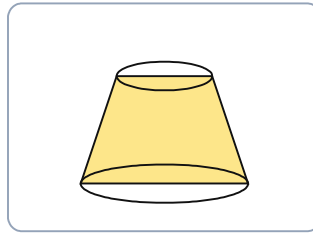
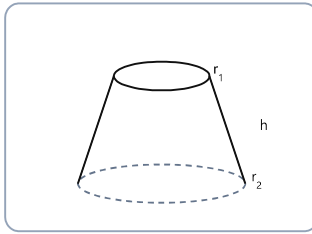
- **Total surface area (with base):**

$$SA = 3\pi r^2$$

- **Volume:**

$$V = \frac{2}{3}\pi r^3$$

### Frustum of a Right Circular Cone

top radius  $r_1$ bottom radius  $r_2$ height  $h$ slant  $s$ 

- Slant height:**

$$s = \sqrt{h^2 + (r_2 - r_1)^2}$$

- Lateral area:**

$$LA = \pi(r_1 + r_2)s$$

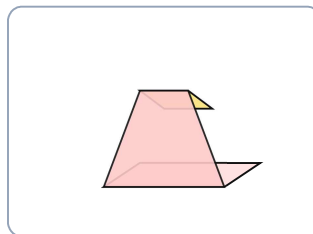
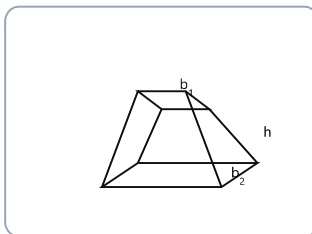
- Surface area:**

$$SA = LA + \pi r_1^2 + \pi r_2^2$$

- Volume:**

$$V = \frac{1}{3}\pi h(r_1^2 + r_1 r_2 + r_2^2)$$

### Frustum of a Right Regular Pyramid

top base area  $A_1$ , perimeter  $P_1$ bottom base area  $A_2$ , perimeter  $P_2$ height  $h$ slant  $s$ 

- Lateral area (regular):**

$$LA = \frac{1}{2}(P_1 + P_2)s$$

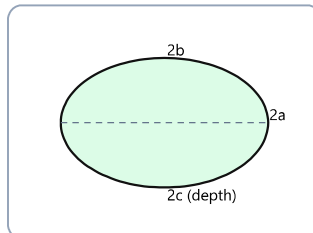
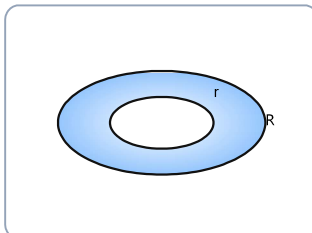
- Surface area:**

$$SA = LA + A_1 + A_2$$

- Volume (general):**

$$V = \frac{1}{3}h(A_1 + A_2 + \sqrt{A_1 A_2})$$

### Torus & Ellipsoid (quick refs)

Torus: major  $R$ , minor  $r$ Ellipsoid semiaxes  $a, b, c$ 

- Torus:**

Surface area and volume (donut with tube radius  $r$  around circle radius  $R$ )

$$SA = 4\pi^2 Rr$$

$$V = 2\pi^2 Rr^2$$

- Ellipsoid:**

$$V = \frac{4}{3}\pi abc$$

Surface area requires elliptic integrals; a common approximation (Knud Thomsen) uses exponent  $p \approx 1.6075$ :

$$SA \approx 4\pi \frac{a^p + b^p + c^p}{3}$$

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